

SE Lab 1 – Requirements Engineering & UML Use-Case Modelling

Problem Statement #11: Telemedicine Slot Booking & Prescription Portal

Healthcare & Telemedicine

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1. Requirements Table

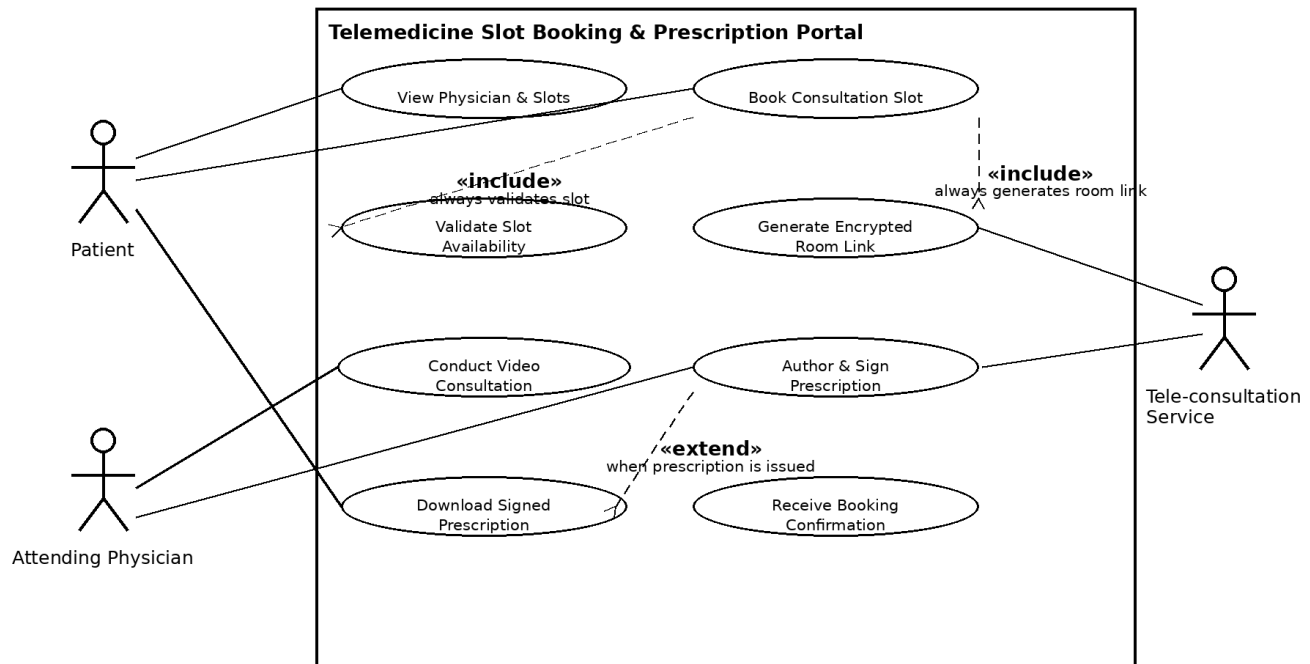
Exactly five Functional Requirements (FR-001 to FR-005) and two Non-Functional Requirements (NFR-001 to NFR-002), based on Problem Statement #11 and the Lab 1 handout.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable attending physicians to author and digitally sign structured prescriptions that are automatically compiled into downloadable PDF records.	High	Pass: A prescription PDF contains the validated physician identity/license information and a valid digital signature. Fail: A prescription is generated without physician license validation or signature.	Supports the core prescription-portal function and ensures that digital prescriptions are attributable to the attending physician.
FR-002	Functional	The system shall allow patients to view available physicians and their specialties and select an available video consultation slot.	High	Pass: The system displays the selected physician's specialty and only allows selection of an available slot. Fail: An unavailable slot can be booked.	Enables patients to discover suitable physicians and choose a valid consultation time.
FR-003	Functional	The system shall allow patients to book a selected video consultation slot and generate a unique booking record containing the patient, physician, date, and time.	High	Pass: A successful booking receives a unique booking ID and stores patient, physician, date, and time. Fail: A booking is created without any required booking field.	Provides the central scheduling capability of the telemedicine portal.
FR-004	Functional	The system shall generate and deliver an encrypted tele-consultation room link to the patient and attending physician for each confirmed booking.	High	Pass: Each confirmed booking receives a unique consultation-room link accessible only to the authorized patient and physician. Fail: A booking has no room link or the link is exposed as unprotected/plain room information.	Enables secure remote consultation and reflects the scenario requirement for encrypted tele-consultation room links.
FR-005	Functional	The system shall allow the attending physician to access a patient's confirmed consultation, complete the consultation, and make the digitally signed prescription available to the patient when one is issued.	High	Pass: The physician can access the confirmed consultation and, when a prescription is issued, the patient can download the signed PDF from the booking record. Fail: An issued prescription cannot be linked to the consultation or downloaded by the authorized patient.	Connects consultation completion with the prescription-delivery workflow.
NFR-001	Nonfunctional – Performance & Security	All video session metadata and patient clinical notes must be encrypted in transit using TLS 1.3 and at rest with AES-256.	High	Pass: Security testing verifies TLS 1.3 for data in transit and AES-256 encryption for stored session metadata and clinical notes. Fail: Either required protection is absent.	Protects sensitive telemedicine and clinical information and follows the security requirement provided in the problem statement.
NFR-002	Nonfunctional – Performance & Availability	The system shall display available consultation slots and complete a slot-booking request within 3 seconds under the defined simulated peak-load test.	Medium	Pass: In load testing, at least 95% of slot-view and booking requests complete within 3 seconds without creating duplicate bookings. Fail: The 95% target is not achieved or duplicate bookings occur.	Provides responsive booking during periods of high demand while protecting scheduling consistency.

2. UML Use-Case Diagram

The problem statement explicitly identifies Patient and Attending Physician as target actors. The Lab 1 handout asks for at least three actors, so a Tele-consultation Service is modeled as a supporting external actor inferred from the requirement for encrypted tele-consultation room links.

UML Use-Case Diagram - Telemedicine Slot Booking & Prescription Portal



Actors: Patient, Attending Physician, Tele-consultation Service (supporting external service inferred from the encrypted room-link requirement).

3. Use-Case Flow Specification

Use Case: Book Video Consultation Slot

Preconditions

- The Patient is authenticated in the Telemedicine Slot Booking & Prescription Portal.
- Attending physicians and their available consultation slots are registered in the system.
- The selected physician has at least one available slot.
- The telemedicine booking service is available.

Postconditions

- A unique booking record is created for the selected patient, physician, date, and time.
- The selected slot is marked as booked and is no longer available for another patient.
- An encrypted tele-consultation room link is generated and associated with the confirmed booking.
- The patient and attending physician can access the consultation details.

Main Success Scenario

1. Patient selects a physician or specialty to view available consultation slots.
2. System displays the physician's available video consultation slots.
3. Patient selects a preferred available date and time.
4. System validates that the selected slot is still available.
5. System creates a unique booking record containing the patient, physician, date, and time.
6. System marks the selected slot as booked.
7. System generates an encrypted tele-consultation room link.
8. System associates the room link with the booking and makes it available to the patient and attending physician.
9. System displays the booking confirmation and consultation details to the patient.
10. Use case ends successfully.

Alternate Flow

4a. Selected slot is no longer available: The system informs the patient that the slot has been taken and displays the physician's currently available slots.

4a1. Patient selects another available slot.

4a2. The system validates the new slot and resumes at Step 5.

7a. Room-link generation fails: The system does not confirm the consultation-room access as ready; it records the failure and prompts the patient to retry or select another available slot after the booking state is safely handled.

Actors Involved

Primary Actor: Patient

Supporting Actor: Attending Physician

Supporting External Actor: Tele-consultation Service